

Themed plots

`themed` wraps a plot function to render in both light and dark modes, producing a single HTML element that switches on `prefers-color-scheme`. The same function runs twice – once per theme – so you can use `light_dark()` inside to pick theme-dependent values.

```
import matplotlib.pyplot as plt
import numpy as np

from mini.lit import memo
from mini.vis import themed
from mini.vis.theme import light_dark

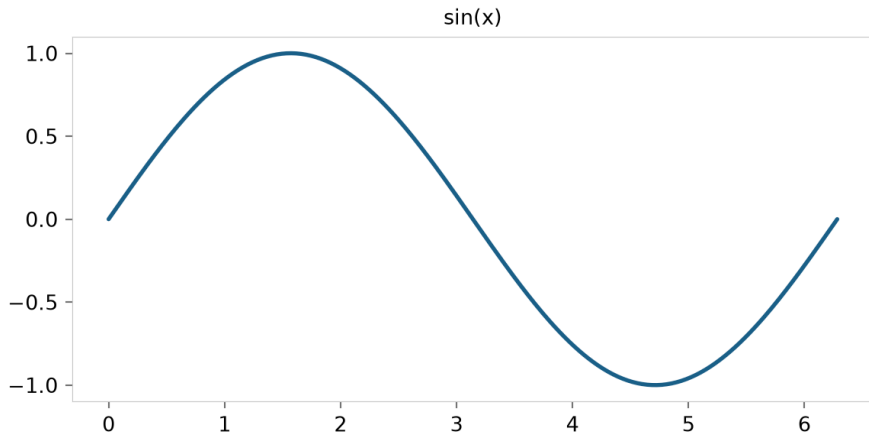
x = np.linspace(0, 2 * np.pi, 300)
```

Plain decorator

The simplest form: `@themed` with no arguments. A cell's last expression is displayed, and `themed` returns the figure's HTML, so the figure appears here.

```
@themed
def plot_plain() -> plt.Figure:
    fig, ax = plt.subplots(figsize=(6, 3))
    ax.plot(x, np.sin(x), color=light_dark("#1a5f8a"),
    ax.set_title("sin(x)")
    return fig
```

```
plot_plain()
```

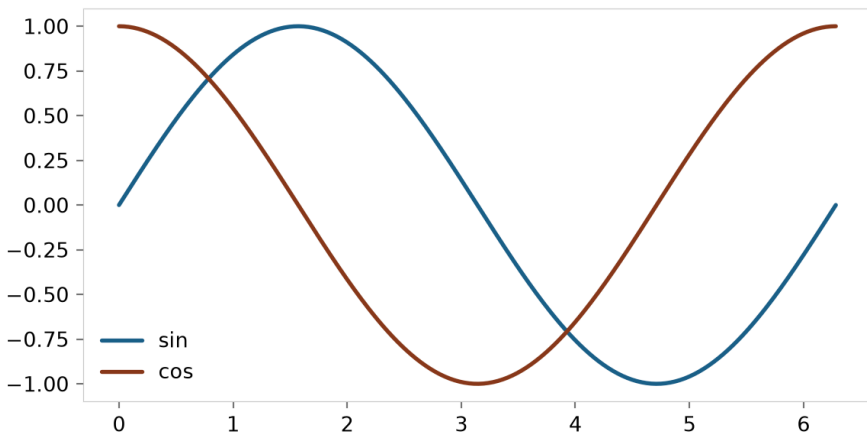


Decorator factory

Pass keyword arguments to set `alt_text`, `caption`, `max_width`, or custom styles. This is the form you want when defining a standalone plot function. Stacking `@memo` outside it caches the rendered HTML (and keeps the two PNGs it wrote) keyed by the function's source and arguments, so re-rendering this page after a prose edit does not redraw the figure.

```
@memo
@themed(alt_text="sin and cos", caption="Two sinusoids", max_width=600)
def plot_factory(x: np.ndarray) -> plt.Figure:
    fig, ax = plt.subplots(figsize=(6, 3))
    color_sin = light_dark("#1a5f8a", "#6ab0d4")
    color_cos = light_dark("#8a3a1a", "#d49a6a")
    ax.plot(x, np.sin(x), color=color_sin, lw=2, label="sin")
    ax.plot(x, np.cos(x), color=color_cos, lw=2, label="cos")
    ax.legend()
    return fig
```

plot_factory(x)



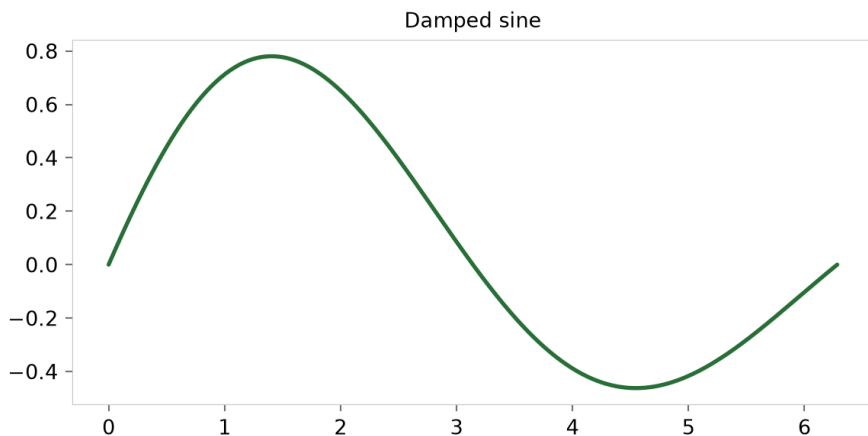
Two sinusoids, a quarter period apart.

Direct call

Useful for one-off plots, or when wrapping a function defined elsewhere.

```
def _plot_raw() -> plt.Figure:
    fig, ax = plt.subplots(figsize=(6, 3))
    ax.plot(x, np.sin(x) * np.exp(-x / 6), color=light
    ax.set_title("Damped sine")
    return fig

themed(_plot_raw, alt_text="Damped sine wave")()
```



The prose can also quote values from the namespace: the grid has 300 points and its last value is 6.283.